

Artificial Intelligence for Corporate Finance

AI-Augmented Valuation: Delegation, Scenarios and Reconciliation

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Structure of the course

- **Email:** amedeo.andriollo@dauphine.psl.eu
- **Assessment:** 150-minute final in-class examination.
- **Office hours:** feel free to (DO) come: B612.
- Slides/materials are updated regularly. Report mistakes: corrections help the whole class.
- **This session:** delegate valuation work to AI without delegating the valuation *judgement*.
- **Built on:** Rosenbaum & Pearl, *Investment Banking*; Koller, Goedhart & Wessels, *Valuation*.

Session 3: the case assignment

Today's task:

How to delegate a valuation to AI w/out allowing to substitute my judgment?
(=silently define comparability/normalize earnings/forecasting/reconciliation)

Session 1 Do not take for granted AI: what to delegate is key.

Session 2 Do not take for granted data: check consistency.

Session 3 Do not take for granted the plausibility of *the valuation* from the AI.

Again, the same problem: who is taking the decision?

Our route: profile → audited model → valuation

Output from Session 1

Business profile

Products, customers, operations, and unresolved questions.

Output from Session 2

Audited historical model

FY2025A revenue EUR 481.0m;
reported EBITDA EUR 68.8m;
adjusted scenario EUR 70.2m.

Session 3

Preliminary sell-side range

comparables analysis + precedent transactions + DCF + valuation reconciliation.

Output of this session: a preliminary range for the Enterprise Value, reconciliation from Enterprise Value to Equity Value, base case DCF, and unresolved valuation issues.

Refresh: three valuation methodologies

Method	Core question
Comparable Companies	At what multiples are comparable publicly traded companies trading, and what valuation range do those multiples imply for the target?
Precedent Transactions	What multiples were paid in comparable prior M&A transactions, and what valuation range do those multiples imply for the target?
DCF	What enterprise value is implied by the present value of projected unlevered free cash flow and terminal value?

Implied equity value

Implied Equity Value
= EV - Net Debt.

The three valuation methodologies use **different information sets** and should not be averaged.

Valuation in a Sell-Side M&A Process

Organization and Preparation (before)

- establish preliminary valuation expectations for the seller;
- inform sale-process design and buyer targeting;
- identify key valuation assumptions requiring further diligence.

During the Sale Process

- evaluate bids and offers against the preliminary valuation;
- evaluate bids based on price, transaction terms and execution risk;
- (..negotiation).

**A preliminary valuation range supports the sale process;
it is not a prediction of the eventual transaction price.
(it works also the other way round!)**

Calculations vs Valuation

	Tasks that can be delegated to AI	Valuation judgments that require approval
Comparable Companies	Calculate trading multiples and statistics; apply selected multiples to the target	Define the peer-selection criteria (i.e., the comparables), and determine the appropriate treatment of EBITDA adjustments.
Precedent transactions	Calculate transaction multiples and the implied valuation range	Select the comparable acquisitions and assess transaction-specific considerations (timing and market conditions).
DCF	Project free cash flow, DCF, calculate terminal value, and run sensitivity analysis	forecast assumptions, treatment of non-recurring items, and assumptions underlying continuing value.
Reconciliation	compare the valuation	Determine the preliminary valuation range and articulate the rationale for that conclusion.

Calculations may be delegated (AI); valuations must remain explicit/accountable (Human).

Define the AI's role before writing the prompt

Order of tasks

Required valuation deliverable → valuation judgments
→ tasks delegated to AI → What the AI is permitted to conclude
→ recollect source information/evidence → defined independent verification → prompt

Central sentence

Do not write the prompt until you have identified
i) which valuation judgments remain with the analyst and
ii) what the AI is permitted to do.

Why: Requests such as *“select the comparable companies”*, *“determine the assumptions for the DCF”* and *“reconcile the valuations”* implicitly delegate valuation judgments
↪ unless the relevant **selection criteria and constraints** are defined in advance.

Delegation Canvas: decisions before the prompt

1. Required (valuation) deliverables.
 2. Valuation judgments requiring analyst decision.
 3. What to delegate to AI.
 4. Conclusions the AI may draw.
 5. What AI is not allowed to conclude (i.e., conclusions reserved to the analyst).
 6. Financial data, Company info, assumptions and other sources to be supplied to the AI.
 7. Independent verification procedure.
 8. Conditions requiring the task to stop for analyst review.
 9. Allocation of the six-message budget (prompt budgeting)
 10. Initial prompt, written in full (character budgeting).
- Closed-laptop sequence:** decompose the valuation main task into subtasks, identify the judgments retained by the analyst, define the source information and independent verification, then write the prompt precisely.

CLOSED LAPTOP — 25 minutes

Now

No ChatGPT/Claude. No laptop.

1. Read the valuation assignment.
2. Complete the Delegation Canvas on paper.
3. Plan how you will use the six analytical AI messages.
4. Write the exact first prompt word for word.

**Your first deliverable is a delegation design, not an AI answer:
learn to create an analytical architecture before seeing what an LLM suggests.**

Compare AI-use plans and the six-message analytical sequence

Compare

- Which valuation judgments remain with the analyst?
- What conclusions may the AI draw?
- What source information or evidence is supplied for each task (trade-off with independent verification)?
- What conditions require the task to stop for analyst review? (ambiguities: return to analyst, not silently decide).

Constraint

Up to six analytical AI messages.
Did you use all of them?

Good delegation reduces wasteful prompting:

prompt count is not the lesson, disciplined decomposition of the valuation analysis is.

Valuation info across multiple sources

Source	What it contributes
Company profile (s01)	products, customers, route to market, operations, and management-provided information;
Historical spreadsheets (s02)	audited FY2025A historical financial information, including reported EBITDA and management-adjusted EBITDA (68.8m vs. 70.2m);
Valuation Deck (today)	candidate comparable-company info, management's comparable-company classifications, and information on prior M&A transactions;
Management Case (today)	management's 2026E–2030E forecast assumptions/rationale;
Valuation workbook (today)	source data, calculation framework

Each AI task should identify the source materials relevant to that task because no single source contains all the information required for the valuation analysis.

↪ why do we wish to keep the sources separated?

Excessive delegation in performing comparables analysis

Prompt

"Here are the candidate public companies. Select the best comparable companies for TargetCo and value TargetCo using EV/Revenue and EV/EBITDA."

Plausible AI response

- 1) Selects *management-designated* peers,
- 2) relies on *sector classifications* as evidence of comparability,
- 3) uses an *average trading multiple* to summarize, and presents an implied valuation range.

Valuation judgments implicitly delegated to AI

1. criteria used to assess comparability, and the relative importance of those criteria;
2. which candidate companies to include/exclude; treatment of borderline comparables;
3. whether the available information is sufficient;
4. whether to use reported vs. management-adjusted EBITDA;
5. which summary statistic.

The prompt implicitly delegates substantive valuation judgments to AI.

Assess the evidence before selecting comparable companies

Stage 1 — AI extracts the relevant source information

Candidate company | Products | End markets | Business model | Geography | Size | Revenue growth | EBITDA margin | ROIC | CapEx/Revenue and D&A | Source and info limitations

Constraint

Do not select the final comparable-company set. Do not create an aggregate comparability score or ranking. Do not calculate an implied valuation range at this stage.

↪ **Stage 2:** Comparability criteria: AI may apply those criteria and identify borderline candidates. For instance, *is Tessera a comparable company?*

The evidences are mixed:

- Ceres has an attractive packaging/healthcare label, but materially higher growth, margin and ROIC.
- Alpina has a superficially relevant rigid-packaging label, but a different capital-investment model.
- Tessera lacks the packaging industry label, yet similar qualification, component-manufacturing and capital economics.

Assess comparability using business and financial characteristics

Dimension	Question	Possible role ¹
Products, customer/ end-market exposure	<i>Are the products, customers, and end-markets sufficiently similar to generate comparable operating economics?</i>	Primary
Business model and route to market	<i>Are manufacturing model, distribution structure, and sales channels comparable?</i>	Primary
Customer qualification and switching characteristics	<i>Are qualification requirements, tooling commitments, design-in processes, and switching costs comparable?</i>	Primary / secondary
Geographic exposure and company size	<i>Are the relevant markets, operating footprint, and company size sufficiently comparable?</i>	Secondary
Revenue growth, EBITDA margin, and ROIC	<i>Do differences in growth, profitability, and returns help explain differences in trading multiples?</i>	Diagnostic
D&A and capital expenditures	<i>Do differences in depreciation and capital expenditures indicate materially different capital intensity or reinvestment requirements?</i>	Diagnostic / exclusion
AI can operationalize the comparability rule, but the analyst must define the rule and the hierarchy of dimensions.		

¹The hierarchy is case-specific and should be justified by the analyst.

Comparables-Company Universe: 14 candidates

- **Nordwell** Closures
- **Luminex** Dispensing
- **Veridia** Rigid Solutions
- **Helix** Consumer Dispense
- **Ceres** Pharma Pack
- **Marell** Food Containers
- **Alpina** Glass and Metal
- **Solv** Healthcare Components
- **Flexora** Films
- **Baltic** Molded Fiber
- **Kernhaus** Distribution
- **Orbis** Recycled Materials
- **Tessera** Precision Components
- **Vela** Contract Filling

Management-designated core peers

Company	Management rationale	Key comparability issue to assess
Nordwell	Product overlap, European footprint and direct/key-account selling	Confirmation of end-market mix, qualification requirements and operating economics.
Luminex	Dispensing exposure, qualification intensity and key-account business model	Confirmation of product/end-market mix and financial comparability.
Ceres	TargetCo's growing Healthcare exposure and Ceres's regulated packaging capabilities	Test whether TargetCo's healthcare mix, growth, margins and ROIC are sufficiently comparable.
Alpina	Rigid packaging formats, premium end markets and direct brand-owner relationships	Test whether different production technology, D&A, CapEx and ROIC make its economics insufficiently comparable.

Question:

Does the management-designated comparable reflect TargetCo's current business and financial characteristics?

(Or primarily the economics management expects TargetCo to achieve?)

↪ Management's proposed comparable-company set is an input to the analysis, not a substitute for (independent) criterion.

AI-organized representation of useful inputs

Candidate company	Business model	Revenue growth	ROIC	CapEx/ Revenue	Relevant information not supplied / comparability limitation
Nordwell	Technical component manufacturer	4.2%	13.6%	5.4%	Healthcare mix not supplied.
Ceres	Regulated pharmaceutical-packaging specialist	6.2%	20.8%	5.0%	TargetCo's regulated-pharma mix and economic profile are not shown to be comparable.
Alpina	Glass and metal packaging manufacturer	3.0%	9.4%	9.2%	Different production technology and capital-investment requirements.

Business model, products, customers, qualification: *Is this company sufficiently comparable to include?*
+ growth rate, margin, ROIC, CapEx: *Does this company have sufficiently similar economics to include?*

Allowed to be delegated: AI may extract source info, organize comparisons, and identify contradictions or missing info.

Not allowed to delegation: the analyst defines the comparability criteria and approves the comparable-company set.

Students: define the criteria, then select the comparables

1. Identify 3/4 **comparability criteria** you consider most important for TargetCo and justify them (e.g. product/customer similarity).
2. Identify additional characteristics that will be used as **supporting** evidence or **diagnostic** checks (e.g., geography may be supporting evidence).
3. State any (at least one) characteristics that would make a candidate **unsuitable** as comparables (e.g., distributor rather than a manufacturer).
4. a) Select the main comparable companies (e.g., low/high) + identify any additional companies retained for sensitivity analysis/cross-checking (i.e., borderline cases).
5. a) Ask AI to apply the analyst-defined criteria + b) identify candidates for which i) the criteria point in different directions or ii) the available information is insufficient.

False precision test

If AI reports that a company is “8.4/10 comparable,” ask: *which comparability criteria were used, how were they measured and weighted, how was missing information treated, and what rule maps the resulting score into inclusion or exclusion?*

↪ **A score may organize a judgement for which you will be accountable.**

Try yourself - 25 minutes

1. State your comparability criteria and any exclusion conditions
2. Select the companies for your principal Comparable Companies Analysis and identify any additional companies retained for cross-checking.

Now

Allow to use AI

3. Ask the AI to apply your criteria and identify conflicting or missing information.

Be ready to defend one inclusion and one exclusion!

Debrief: "comparing" comparables

Comparables	Members	Interpretation
Selected	Nordwell, Luminex, Veridia, Helix	Strong similarity in products, customer/end-market exposure, business model, qualification characteristics, growth, profitability, ROIC and capital-investment requirements.
Alternative set (out-of-industry)	main suspects + Tessera	Tessera is outside the packaging industry classification but has similar component-manufacturing, qualification, growth, ROIC and capital-investment characteristics.
Management-designated	Nordwell, Luminex, Ceres, Alpina	Includes two close operating comparables, one company with materially stronger regulated-pharma growth/ROIC economics, and one with materially different production technology and capital intensity.

No single truth: different comparable-company sets may coexist, but use explicit defensible (non-AI-delegated) criteria.

Revenue growth, profitability and ROIC help interpret multiple differences

Candidate company	Revenue growth	EBITDA margin	ROIC	EV/Revenue	EV/EBITDA
Alpina	3.0%	15.0%	9.4%	1.17x	7.81x
Veridia	3.8%	14.1%	12.6%	1.11x	7.85x
Nordwell	4.2%	14.8%	13.6%	1.13x	7.63x
Tessera	4.5%	15.2%	13.7%	1.17x	7.70x
Helix	4.7%	15.6%	14.4%	1.21x	7.76x
Luminex	5.6%	16.5%	15.2%	1.22x	7.40x
Ceres	6.2%	18.9%	20.8%	1.74x	9.17x

Interpretation

Companies with similar products can trade at different multiples when their revenue growth, profitability and ROIC differ materially.

$$\frac{EV}{Revenue} = \frac{EV}{EBITDA} \times EBITDA \text{ Margin.}$$

Next question: Alpina trades at an EV/EBITDA multiple similar to the selected comparable companies despite materially lower ROIC. What does EV/EBITDA fail to capture?

EV/EBITDA may obscure differences in capital intensity

Company	EV/EBITDA	EV/EBIT	EV/NOPAT	CapEx/Revenue	ROIC
Veridia	7.85x	11.64x	15.53x	5.8%	12.6%
Nordwell	7.63x	10.85x	14.47x	5.4%	13.6%
Helix	7.76x	11.02x	14.69x	5.4%	14.4%
Luminex	7.40x	10.19x	13.59x	5.2%	15.2%
Alpina	7.81x	13.89x	18.52x	9.2%	9.4%

Interpretation

Alpina (management-selected) trades at an EV/EBITDA multiple similar to the selected comparable companies, but its higher depreciation and capital expenditures and lower ROIC indicate materially different capital intensity.

The criteria:

- **Primary multiples:** EV/Revenue and EV/EBITDA.
- **Additional diagnostics:** EV/EBIT, EV/NOPAT, CapEx/Revenue and ROIC help identify differences in depreciation, capital intensity and reinvestment requirements that EV/EBITDA does not capture.

TargetCo FY2025A EBITDA: reported vs. management-adjusted bases

Reported FY2025A EBITDA

EBITDA = EUR 68.8m

Audited FY2025A result

Management-adjusted FY2025A EBITDA

$68.8 + 1.4 = \text{EUR } 70.2m$

Management proposes a EUR 1.4m adjustment for external commissioning support and duplicate QA costs.

Do not create an unsupported intermediate EBITDA.

Keep the EBITDA basis explicit:

Do not average or combine the two measures without analytical support.

↪ comparables multiples are fixed, TargetCo's EBITDA basis is the only variable.

Comparable Companies Analysis: implied EV

TargetCo financial statistic	Low selected multiple	Median selected multiple	High selected multiple
FY2025A Revenue: EUR 481.0m	1.11x → 533.3	1.17x → 562.5	1.22x → 585.7
Reported FY2025A EBITDA: EUR 68.8m	7.40x → 509.3	7.69x → 529.3	7.85x → 539.8
Management-adjusted FY2025A EBITDA: EUR 70.2m	7.40x → 519.6	7.69x → 540.1	7.85x → 550.8

Implied Enterprise Value ranges: EV/Revenue: 533.3-585.7m; EV/EBITDA: 509.3-539.8m VS. 519.6-550.8m.

Effect of the management adjustment

At the median selected EV/EBITDA multiple, accepting the EUR 1.4m management adjustment increases implied Enterprise Value by approximately EUR 10.8m:

$$7.69 \times \text{EUR } 1.4\text{m} \approx \text{EUR } 10.8\text{m}.$$

Precedent Transactions Analysis uses a different information set

Comparables

Current trading multiples of selected publicly traded comparable companies.

Precedent transactions

Purchase-price multiples observed in selected prior M&A transactions involving comparable companies.

Assess **transaction comparability** using the target business, transaction date, market and industry conditions, buyer type, transaction structure, process information and special circumstances.

Observed transaction multiples do not **by themselves** identify the causal contribution of synergies, scarcity, competitive tension or seller circumstances to the purchase price.

Student work - 15 minutes

Now

Use one AI prompt for this stage

1. Ask AI to organize the supplied transaction evidence into a comparison table. Include the relevant multiples, business comparability, circumstances.
2. In your prompt, ask the AI to: i) distinguish facts from causal explanations that are not established by the evidence; and ii) refrain from selecting the transactions for you.
3. Using the AI output and the source materials, decide which transactions to include. Record your selection and the principal caveat for each transaction.
4. Independently verify EV/LTM Revenue and EV/LTM EBITDA in Excel, and record the AI prompt and how its output was used in the AI Delegation Log.

The (accountable) deliverable is the analyst's selection.
AI may organize the evidence but the analyst determines:
i) which transactions enter and ii) documents how the AI output was used.

Precedent Transactions Analysis: 8 candidate transactions

Target	Announce date	EV/LTM Revenue	EV/LTM EBITDA	Transaction facts
Capella Closures	Jun-24	1.19x	8.16x	European technical-closure manufacturer; EUR 24m annual procurement and cross-selling plan announced.
LumiDose	Sep-23	1.23x	7.90x	Dispensing-components supplier; accelerated sale process following working-capital disruption and covenant pressure.
Veralux Pack	Nov-22	1.14x	8.06x	Specialty rigid-packaging and closures business; transaction occurred during a capacity-expansion ramp and demand-normalization period.
PharmaNest	Apr-25	1.85x	8.09x	Primary pharmaceutical-packaging specialist; announcement described it as one of few independent European suppliers qualified for two regulated delivery formats.
Northbay	May-21	0.70x	7.29x	2021 transaction involving a standard food-container business with limited technical differentiation; followed a year of weak food-service demand.
FlexiSeal	Feb-24	1.07x	7.79x	Flexible-film and pouch manufacturer; EUR 30m procurement and footprint synergy plan announced.
EuroPack	Mar-23	0.67x	9.09x	Pan-European packaging distributor; carve-out transaction; standalone-cost adjustments not quantified.
MediCore	Aug-25	2.67x	N/A ²	Medical-device components supplier with proprietary IP; comparable LTM EBITDA unavailable.

**Do not select transactions based on the highest multiples or the most recent dates.
Assess business comparability, timing, transaction-specific circumstances and denominator quality.**

²N/A because comparable LTM EBITDA is not supplied.

Debrief: a prompt that converts facts into unsupported claims

Audit your own prompting choice: what analytical judgment has been delegated to AI?

Prompt

“**Explain why** the candidate transactions with the highest observed EV/LTM EBITDA multiples were priced at higher multiples than the other transactions.”

Plausible weak response

“The higher transaction multiples likely reflect strategic fit, meaningful synergy potential, scarcity of differentiated assets and competitive bidding tension.”

What the evidence does establish:

- Certain buyers announced synergy plans (disclosed for only two transactions).
- One transaction description contains evidence consistent with scarcity.
- Some transactions contain seller/process circumstances.
- Buyer type and transaction context are supplied.

What the evidence does not establish:

- how synergies affected the purchase price;
- whether competitive bidding affected price;
- whether scarcity caused a higher multiple;
- whether “strategic fit” explains any observed multiple difference.

↪ Relevant facts $\neq$ cause of purchase prices

The prompt delegates causal interpretation that the available transaction evidence cannot support.

Transaction evidence: what I would want from the AI.

Better question

Which supplied facts are relevant to interpreting and which conclusions are not established?

Supplied fact	What the evidence supports	What the evidence does not establish
Buyer announced a EUR 24m annual procurement and cross-selling plan (Capella).	The announced synergy plan is a transaction-specific fact that may be relevant when interpreting the buyer's economics.	The synergy plan caused the observed purchase price or transaction multiple, or how much value was attributable to it.
Sale process accelerated after ERP-related working-capital disruption and a covenant test (LumiDose); no insolvency filing occurred.	A transaction-specific seller/process circumstance that should be considered when assessing comparability.	A quantified "distress discount" or quantify any effect on the purchase price.
PharmaNest as one of few independent European suppliers qualified for two regulated delivery formats.	Relevant info about the target's differentiation and market position.	A quantified scarcity premium, competitive bidding tension, or that scarcity caused the observed multiple.
Comparable LTM EBITDA is not supplied (MediCore).	EV/LTM EBITDA cannot be calculated from the supplied information.	The missing EBITDA should not be inferred, estimated or reverse-engineered w/out additional evidence.

↪ Transaction facts are supporting info (not explanation).

Debrief: selecting transactions

Transaction	Selection status	Analyst rationale / principal caveat
Capella Closures	Include	Strong business comparability. The buyer announced a EUR 24m synergy plan, but the supplied evidence does not establish its effect on the purchase price.
LumiDose	Include	Strong business comparability. The accelerated sale process following working-capital disruption and a covenant test is a transaction-specific caveat; the evidence does not establish a quantified distress discount.
Veralux Pack	Include	Strong business comparability. The transaction is older, but its capacity-expansion and demand-normalization context is relevant when assessing comparability.
PharmaNest	Contextual reference	Regulated pharmaceutical-packaging business with materially different product mix and differentiation. The announcement describes the target as one of few qualified independent suppliers, but no scarcity premium or bidding effect is identified.
Northbay	Exclude	Older transaction involving a more standardized food-container business with limited technical differentiation and weaker business comparability.
FlexiSeal	Exclude	Flexible-film and pouch production differs materially from TargetCo's component-manufacturing model.
EuroPack	Exclude	Distributor rather than manufacturer; carve-out transaction with standalone-cost adjustments not quantified in the supplied evidence.
MediCore	Exclude	Different medical-device product scope and proprietary-IP economics; comparable LTM EBITDA is unavailable.

Precedent Transactions Analysis: implied EV and limits

TargetCo financial statistic	Low selected multiple	Median selected multiple	High selected multiple
FY2025A Revenue: EUR 481.0m	1.14x → 547.0m	1.19x → 573.5m	1.23x → 589.2m
Reported FY2025A EBITDA: EUR 68.8m	7.90x → 543.7m	8.06x → 554.2m	8.16x → 561.3m
Management-adjusted FY2025A EBITDA: EUR 70.2m	7.90x → 554.8m	8.06x → 565.5m	8.16x → 572.7m

What it tells us:

1) The selected EV/LTM EBITDA transaction multiples are above the selected comparable-company EV/EBITDA multiples, while 2) the EV/Revenue multiple ranges overlap substantially.

What it does not tell us:

The analysis does not predict TargetCo's eventual transaction price, identify a standalone control premium, or establish why any individual transaction occurred at its observed multiple.

Any difference between precedent-transaction and comparable-company multiples are observed valuation evidence.

BREAK — 15 minutes

Open the workbook to 07_DCF.

Forecast assumptions require analyst judgment

DCF mechanics

Revenue $\rightarrow$ EBITDA $\rightarrow$ EBIT $\rightarrow$ NOPAT $\rightarrow$ FCFF

$$\text{FCFF}_t = \text{NOPAT}_t + \text{D\&A}_t - \text{CapEx}_t - \Delta\text{NWC}_t$$

$$\text{Enterprise Value} = \sum_{t=1}^T \frac{\text{FCFF}_t}{(1 + \text{WACC})^t} + \frac{\text{Terminal Value}_T}{(1 + \text{WACC})^T}$$

AI can support

- calculate FCFF from stated assumptions;
- apply discount factors and terminal-value formulas;
- run sensitivities;
- calculate implied operating and valuation outputs.

Analyst judgment

- revenue growth and EBITDA margins;
- capital expenditures and working-capital requirements;
- treatment of historical adjustments;
- consistency of forecast ROIC and reinvestment;
- continuing-value assumptions and steady-state consistency.

The delegation question is not whether AI can calculate a DCF. It is i) which (forecast) assumptions AI may analyze and ii) which valuation judgments the analyst must make/verify.

Management Case and Analyst Base Case

Management Case

Management's complete 2026E–2030E forecast is supplied as the starting case.

Evaluate the forecast assumptions: first the rationale, then the individual assumptions.

Base Case

Develop a base case by changing only a limited number of (clearly identified) forecast drivers. For each change, record:

- the Management assumption;
- the Base assumption;
- the supporting evidence/rationale.

Primary forecast drivers: i) capacity utilization and revenue growth, ii) EBITDA-margin path, iii) capital expenditures and reinvestment requirements, and, where supported by the evidence, iv) operating NWC as a percentage of revenue.

Each change should respond to an identified issue and remain transparent+independently reviewable.

An underspecified prompt can manufacture a Base Case

Prompt

“Build a reasonable Base Case using management’s assumptions and the historical model.”

Plausible AI response

“I reduced management’s revenue-growth assumptions by 0.5 percentage points, set the terminal EBITDA margin at 17.0%, kept CapEx close to management’s forecast and assumed modest working-capital improvement. The resulting Enterprise Value is approximately EUR 630m, providing a balanced Base Case.”

AI choice	What is missing	Why it matters
Revenue growth –0.5 pp	Evidence supporting the size/timing	Changes revenue, margins, FCFF
Terminal EBITDA margin 17.0%	Operating evidence supporting 17.0% rather than management’s path	Changes terminal FCFF
CapEx “close to management”	Explicit reinvestment requirement consistent with the forecast	Affects FCFF, invested capital and the sustainability of growth
“Modest” improvement	Defined operating NWC assumption	Changes cash conversion and FCFF
“Balanced” Base Case	A criterion for “balanced” or “reasonable”	A polished conclusion can conceal assumptions

The problem is not necessarily implausibility. The problem is that the prompt gives AI no evidentiary rule.

A Base Case should be constructed from explicit assumption changes supported by evidence, then tested through the resulting FCFF, invested-capital and ROIC paths and continuing-value assumptions.

Management Case: assess the forecast assumptions jointly

Management Case	25A	26E	27E	28E	29E	30E
Revenue growth	–	6.9%	6.4%	6.0%	5.2%	4.4%
Capacity utilization	84%	88%	91%	93%	94%	95%
EBITDA margin	14.3%	15.7%	16.5%	17.1%	17.5%	17.8%
CapEx / Revenue	6.1%	6.0%	5.3%	4.8%	4.6%	4.5%
Operating NWC / Revenue	14.0%	14.0%	13.9%	13.7%	13.5%	13.4%
Implied ROIC	10.6%	12.8%	14.4%	16.0%	17.3%	18.4%

The consistency test

No single forecast path should be assessed in isolation.

In the Management Case, rising utilization and EBITDA margins (↑) occur alongside declining CapEx/Revenue and operating NWC/Revenue (↓).

↪ these assumptions imply ROIC rising from 10.6% in FY2025A to 18.4% in 2030E.

Management Case: the implied ROIC requires validation

TargetCo FY2025A

10.6% ROIC

Selected comparables

14.0% Median ROIC

Management 2030E

18.4% Implied ROIC

FY2025A TargetCo 10.6% → 18.4% Management 2030E

$\Delta\text{ROIC} = +7.8$ pp versus FY2025A and $+4.4$ pp versus the selected-comparable median.

Interpretation

The Management Case does not become implausible simply because its implied ROIC exceeds the comparables.

↪ It requires the analyst to determine whether the forecast improvements in operating performance and capital requirements are *jointly* supported by the evidence.

Student work: verify the economic implications

Now

Independently test the Management Case

1. Recalculate EBITDA, NOPAT, FCFF and operating invested capital for FY2026E–FY2030E.
2. Compute the implied ROIC path using:

$$\text{ROIC}_t = \frac{\text{NOPAT}_t}{(\text{Operating IC}_{t-1} + \text{Operating IC}_t)/2}.$$

3. Compare the FY2030E result with:
 - TargetCo FY2025A ROIC; and
 - the median ROIC of the selected comparable companies.
4. Identify the forecast assumptions that economically account for the change in ROIC over the forecast period.

Your conclusion should not be “the ROIC is high” or “the forecast is aggressive.”
Explain: i) which operating and capital assumptions produce the change in ROIC and ii) whether the supplied evidence supports that combination.

Management Case and Base Case: economic consequences

FY2030E assumption / output	Management Case	Base Case	Economic interpretation
Revenue	EUR 637m	EUR 606m	Slower realization of the forecast growth path.
Capacity utilization	95%	92%	Less complete utilization of installed capacity by FY2030E.
EBITDA margin	17.8%	16.7%	Less complete improvement in operating profitability.
CapEx / Revenue	4.5%	4.7%	More reinvestment retained after the expansion period.
Operating NWC / Revenue	13.4%	13.8%	Less complete improvement in working-capital requirements.
Implied ROIC	18.4%	15.9%	Quantifying different operating/reinvestment assumptions.
DCF Enterprise Value	EUR 699.6m	EUR 600.3m	The forecast changes affect both explicit-period FCFF and continuing value.

What changed?

The Base Case changes a limited number of operating and reinvestment assumptions and recalculates their consequences throughout the DCF.

The EUR 99.3m difference in Enterprise Value is an output of different forecast assumptions.

Historical EBITDA adjustment: avoid double counting

FY2025A historical EBITDA

$$\begin{array}{r} \text{Reported EBITDA} = \text{EUR } 68.8m \\ + \text{Management-proposed adjustment} = \text{EUR } 1.4m \\ \hline \text{Management-adjusted EBITDA} = \text{EUR } 70.2m \end{array}$$

Forecast EBITDA

$$\text{EBITDA}_t = \text{Revenue}_t \times \text{EBITDA Margin}_t$$

The forecast margins already assume that the FY2025A commissioning and duplicate-quality costs do not recur from FY2026E.

Double counting

Forecast EBITDA from the forecast margin + EUR 1.4m additional forecast saving

**The EUR 1.4m historical adjustment is not a separate forecast add-back.
The incremental forecast add-back is 0.**

Continuing value requires a consistent steady state

Perpetuity-growth method

$$\text{Continuing Value}_{2030} = \frac{\text{FCFF}_{2031}}{\text{WACC} - g}$$

Operating steady-state test

Before applying the formula, assess whether the transition from FY2030E to FY2031E and beyond can sustain:

- continuing revenue growth of 2.0%;
- an EBITDA margin consistent with the base case;
- a sustainable return on invested capital;
- sufficient reinvestment to support continuing growth;
- operating working-capital requirements consistent with that growth.

Base case reference points

2030E EBITDA margin: 16.7%

2030E implied ROIC: 15.9%

Continuing growth: g 2.0%

WACC: 9.5%

The FY2030E forecast provides the starting point.

**A continuing value needs to consider the operating assumptions, reinvestment requirements and growth rate
↪ a mutually consistent economic state.**

DCF sensitivity: continuing assumptions materially affect EV

Base Case Enterprise Value sensitivity (EUR m)

WACC \ g	1.5%	2.0%	2.5%
9.0%	598.1	631.9	670.9
9.5%	570.6	600.3	634.2
10.0%	546.3	572.6	602.5

Base Case value composition

PV of explicit-period FCFF	EUR 157.7m
PV of continuing value	EUR 442.6m

Enterprise Value EUR 600.3m

↪ **73.7%** of EV is the PV of continuing value.

Interpretation

At the Base Case WACC of 9.5%, changing continuing growth from 1.5% to 2.5% changes

⇒ EV from EUR 570.6m to EUR 634.2m.

At continuing growth of 2.0%, changing WACC from 9.0% to 10.0% changes

⇒ EV from EUR 631.9m to EUR 572.6m.

Sensitivity analysis shows how dependent the valuation is on assumptions.

The high continuing-value contribution makes the steady-state assumptions a diligence question.

A bad reconciliation prompt creates false comparability

Bad reconciliation prompt

“Combine the Comparable Companies Analysis, Precedent Transactions Analysis and DCF into the most reasonable final Enterprise Value for TargetCo.”

Plausible AI response

The AI reduces each methodology to a single representative value, assigns 40% / 30% / 30% weights and reports one weighted-average Enterprise Value as the “most reasonable” conclusion.

What the prompt silently assumes

Each methodology can be reduced to one point estimate

The three outputs are summable

40% / 30% / 30% are meaningful

Average resolves disagreement

Why that assumption is problematic

The analyses contain different financial-statistic applications, assumption sets and ranges that should not be discarded mechanically.

Comparable Companies Analysis, Precedent Transactions Analysis and DCF draw on different evidence and depend on different valuation assumptions.

The prompt provides no analytical basis for assigning those weights.

Averaging can conceal rather than resolve differences in comparability, forecast assumptions and continuing-value dependence.

Reconciliation is not a voting exercise.

Differences across methodologies should be explained: be accountable of the criteria.

Reconciliation: preserve method-specific evidence and limitations

Method / case	Valuation	Limitation	Recap
Comparables Analysis	EV/Revenue: EUR 533.3-585.7m Reported-EV/EBITDA: EUR 509.3-539.8m Adjusted-EV/EBITDA: EUR 519.6-550.8m	Selected comparable-company set is small; capital intensity differs across candidates; EBITDA basis remains a choice.	Provides current market-based valuation evidence.
Precedent Transactions	EV/Revenue: EUR 547.0-589.2m Reported-EV/EBITDA: EUR 543.7-561.3m Adjusted-EV/EBITDA: EUR 554.8-572.7m	Transaction selection and transaction-specific circumstances matter; observed differences in multiples do not identify their causal source.	Provides transaction-based valuation.
DCF — Management	Enterprise Value: EUR 699.6m	Joint operating, reinvestment and continuing-value assumptions imply 2030E ROIC of 18.4% and require further validation.	Management forecast case.
DCF — Base	Enterprise Value: EUR 600.3m	73.7% of EV is attributable to the PV of continuing value.	DCF under Base Case assumptions.

Preliminary valuation conclusion

Enterprise Value: **EUR 550m–EUR 610m** Minus: net debt of EUR 170m

↔ Equity Value = **EUR 380m–EUR 440m**

Lower bound - EUR 550m: overlap of the Comparables and Precedent Transactions and by the downside of the Base DCF sensitivity.

Upper bound - EUR 610m: close to the Base Case DCF, allowing upside within the DCF sensitivity (well below the Management's).

Final AI role: challenge the conclusion

VP review prompt

"You are the VP reviewing this valuation. Do not rewrite it and do not choose a new valuation. Identify the three analytical choices or assumptions most capable of materially changing the conclusion. For each, state the evidence or analysis needed to resolve it."

Likely challenge	What remains unresolved	Evidence or analysis needed
Comparable-company selection and EBITDA basis	Whether the selected comparables and the use of reported vs. management-adjusted EBITDA materially affect the valuation.	Customer and end-market comparability; support for the EUR 1.4m historical adjustment; sensitivity to alternatives.
Forecast economics and continuing value	Whether the base case operating improvements, reinvestment requirements and continuing assumptions are mutually consistent.	Support for utilization and margin assumptions; maintenance versus growth CapEx; operating NWC evidence; ROIC and reinvestment analysis.
Precedent-transactions	Whether the selected transactions remain appropriate.	Additional process evidence and standalone adjustments.

AI identifies the analytical vulnerabilities and the evidence needed to investigate them. The analyst determines whether that evidence changes the use of a methodology, the valuation range or the preliminary conclusion.

Session 3 handoff: towards Python

AI supported the analysis

- organized supplied evidence;
- applied analyst-specified rules;
- performed calculations and sensitivity analysis;
- compared assumptions and economic consequences;
- identified inconsistencies/unresolved questions;
- challenged the (preliminary) valuation conclusion.

The analyst determined

- comparables and precedent transactions;
- treatment of the historical EBITDA;
- Forecast assumptions;
- continuing-value assumptions;
- reconciliation and valuation.

Session 4

The selected comparable companies and precedent transactions, verified calculations, Management and Base Cases, DCF sensitivities and reconciliation checks become the benchmark to reproduce and extend in Python.

Session 3 establishes the valuation logic and the assumptions that support it.
Session 4 tests whether that analysis can be reproduced transparently and consistently in code.

Reference: sell-side valuation context

- Valuation analysis helps a seller form expectations before a process starts.
- It can inform positioning, buyer selection, evaluation of indications/offers and later negotiation.
- It does not guarantee market clearing value; process structure, terms, financing and buyer-specific economics still matter.
- Session 3 therefore uses a range with evidence and unresolved issues, not a predicted transaction price.

Source basis: RP-IB Ch. 6, PDF pp. 395–470.

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Reference: EV, equity value and method outputs

Item	Meaning	TargetCo
Enterprise value	Value of operations available to capital providers	Output of comps, precedents and DCF
Net debt	Gross debt less unrestricted cash	$198.0 - 28.0 = 170.0$
Equity value	Residual value to shareholders	$EV - 170.0$
Reported EBITDA	Audited denominator before management adjustment	68.8
Adjusted EBITDA	Management-proposed historical scenario	70.2

Control: do not mix an EV multiple with an equity-value denominator; do not hide the EBITDA scenario.

Source basis: RP-IB Chs. 1–3, PDF pp. 39–238; TargetCo audited base.

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Reference: trading-comps mechanics

1. Define the peer rule and screen the universe.
2. Recalculate $EV/Revenue$ and $EV/EBITDA$ from source fields.
3. Inspect distribution: low, median, high and outliers.
4. Apply selected multiples to consistent TargetCo denominators.
5. Preserve reported/adjusted EBITDA scenarios and explain the range.

$$EV_{Target} = \text{Selected multiple} \times \text{Target metric}$$

Session 3 advance: peer selection and denominator treatment are controlled judgements, not preliminary data cleaning.

Source basis: RP-IB Ch. 1, PDF pp. 39–130.

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Reference: peer selection, growth and ROIC

- Industry classification is a starting point, not a sufficient peer rule.
- Management peer lists can contain aspirational companies.
- Growth and ROIC help explain why similar product labels trade at different multiples.
- Business model, mix and reinvestment can dominate superficial sector overlap.

Analyst use

Use growth and ROIC to diagnose multiple differences and challenge selection. Do not turn them into an unexplained composite score.

Source basis: KGW Ch. 18, PDF pp. 545–570.

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Reference: EBITDA multiple and capital intensity

$$EBITA = EBITDA - D\&A \quad NOPAT = EBIT(1 - t)$$

- Similar EV/EBITDA can conceal different D&A, replacement CapEx and asset lives.
- A capital-intensive peer may look inexpensive on EBITDA yet expensive on EBITA/NOPAT.
- CapEx/revenue and ROIC reveal whether the reinvestment burden is similar.
- EV/EBITA and EV/NOPAT are diagnostics here, not a new valuation method.

Source basis: KGW Ch. 18, PDF pp. 545–570.

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Reference: precedent-selection discipline

Question	Why it matters
Business comparability	Product, customer, business model and geography shape economics.
Timing / cycle	Capital markets and industry conditions change observed multiples.
Buyer type	Strategic and sponsor motivations can differ, but labels do not prove price causality.
Special circumstances	Distress, carve-outs or missing standalone costs can distort comparability.
Denominator quality	Missing or non-comparable LTM data can make a multiple unusable.
Output: usable observations with explicit caveats, not a search for the highest multiples.	

Source basis: RP-IB Ch. 2, PDF pp. 131–186.

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Reference: transaction narrative and causality

Evidence status	Permitted statement
Synergy plan disclosed	Buyer announced the plan; it could affect willingness to pay.
Scarcity stated	The target was described as one of few qualified suppliers.
Seller pressure supplied	Timetable and liquidity context may affect interpretation.
Bidder count missing	Do not claim competitive tension.
Price bridge missing	Do not claim the amount of premium caused by any factor.

“Could plausibly affect” is different from “explains.”

Source basis: Instructor-created transaction evidence; no textbook case reproduced.

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Reference: simplified DCF mechanics

$$EBITDA_t = Revenue_t \times Margin_t$$

$$NOPAT_t = (EBITDA_t - D\&A_t)(1 - t)$$

$$FCFF_t = NOPAT_t + D\&A_t - CapEx_t - \Delta NWC_t$$

$$EV = \sum_{t=1}^5 \frac{FCFF_t}{(1 + WACC)^t} + \frac{FCFF_{2031}}{(WACC - g)(1 + WACC)^5}$$

Session boundary: WACC/CAPM derivation is not the lesson; forecast specification and validation are.

Source basis: RP-IB Ch. 3, PDF pp. 187–238.

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Reference: Koller-style model validation

1. **Technical correctness:** formulas, links, signs and units.
2. **Economic consistency:** growth, margins, reinvestment and returns fit together.
3. **Plausibility:** forecast outcomes make sense against history, peers and industry economics.
4. **Scenario / sensitivity:** identify which assumptions drive the conclusion.

Key warning

Individually plausible assumptions can jointly imply implausible ROIC or an incoherent steady state.

Source basis: KGW Ch. 17, PDF pp. 533–544.

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Reference: implied ROIC calculation

$$IC_t = IC_{t-1} + CapEx_t - D\&A_t + \Delta NWC_t$$

$$ROIC_t = \frac{NOPAT_t}{(IC_{t-1} + IC_t)/2}$$

2030E Management example

$NOPAT = 64.0$; average invested capital = 347.1; $ROIC = 64.0/347.1 = 18.4\%$.

Use: translate a forecast of accounting lines into the economic return path those lines jointly imply.

Source basis: KGW Ch. 17, PDF pp. 533–544; Session 3 validator.

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Reference: normalization and forecast double counting

Legitimate role	Not automatic
Assess sustainable FY2025A margin	Add EUR 1.4m to every forecast year
Calibrate confidence in management's margin path	Treat historical adjustment as new incremental saving
Create reported/adjusted trading scenarios	Count the same cost removal in EBITDA margin and add-back

$$\text{Forecast EBITDA} = \text{Revenue} \times \text{Forecast margin} + \underbrace{0}_{\text{separate normalization addback}}$$

Source basis: TargetCo historical normalization rule; Session 3 validator.

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Reference: terminal state, sensitivity and reconciliation

- A terminal growth rate requires reinvestment; growth, ROIC and reinvestment must be mutually consistent.
- High terminal-value share makes the conclusion sensitive to small changes in WACC, g , margin and ROIC.
- Sensitivity identifies diligence priorities; it does not validate the central case by itself.
- Reconciliation compares evidence quality and dominant assumptions. It does not mechanically average outputs.

TargetCo: 73.7% of Base EV comes from terminal value; the 3-by-3 table spans EUR 546.3m–EUR 670.9m.

Source basis: KGW Ch. 17, PDF pp. 533–544; Session 3 validated DCF.

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Reference: completed Delegation Canvas example

Task	AI may conclude	AI may not conclude	Check / stop
Peer evidence matrix	Extract and flag missing/conflicting facts	Final peer set or hidden score	Re-read pack; stop if source is missing
Criteria application	Candidate meets analyst rule	Which dimensions/weights matter	Compare borderline alternatives
DCF diagnostics	Implied ROIC, TV share and sensitivity	"Reasonable" Base assumptions	Recompute FCFF and ROIC
VP challenge	Three material pressure points	New final valuation	Human decides whether range survives

Strong first-prompt pattern

Task + supplied files + allowed conclusion + prohibited conclusion + output schema + evidence citations + stop rule + independent calculation requested.

Source basis: Instructor-designed benchmark; not distributed before the closed-laptop exercise.

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